Keep the mass fixed and change the force

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Proportional / ES: proporcional / PT: proporcional

Double the net force → double the acceleration.

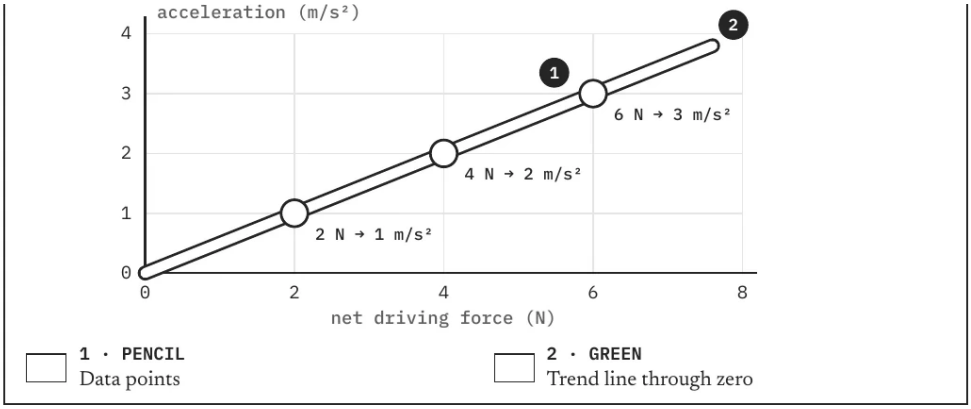
Also on your note page: Driving force, Total moving mass, Friction offset.

B READ THE RELATIONSHIP

Acceleration = net force ÷ total moving mass

As you read, find: driving pull · friction · net force.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

In an ideal 2 kg system, 2 N gives 1 m/s². Predict the acceleration at 4 N with the same mass.

Given F = 2 N → a = 1 m/s² new F = 4 N same mass

Rule a ∝ F (same mass)

Work 4 N ÷ 2 N = 2× the force a = 2 × 1 m/s²

Answer 2 m/s² acceleration doubles

E SAVE FOR THE FINAL MEETING

A bigger net force gives a bigger acceleration, in proportion. That's the rule analysts use to ask how hard XJ would have to be pushed to change course.

Read the physics: Proportional

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Keep the total moving mass the same. With a cart and hanger, move mass from the cart to the hanger. This changes the driving pull while keeping the moving mass constant.

02 HOW IT WORKS

The driving force is one pull. Subtract any opposing friction to find the net force. Divide net force by total mass to find acceleration.

03 WHAT TO CHECK

At fixed mass, doubling the net force doubles acceleration. Doubling the driving pull need not double acceleration when friction is present.

04 TRY THE EXAMPLE FROM YOUR NOTES

In an ideal 2 kg system, 2 N gives 1 m/s². Predict the acceleration at 4 N with the same mass. 2 m/s² acceleration doubles.

05 CONNECT TO XJ

A bigger net force gives a bigger acceleration, in proportion. That's the rule analysts use to ask how hard XJ would have to be pushed to change course.

SAY IT BEFORE YOU WRITE

Which force goes in Newton's second law? The driving pull is ___. After friction, the net force is ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Driving force

ES: fuerza motriz / PT: força motriz

Meaning in my words	How can you tell?
How would you explain driving force to a friend? In my words, this means ___.	A rope pulls right while friction acts left. Which one is the driving force on the cart? I can tell because ___.
Example or sketch	**Non-example and why**
Draw a 2 N driving pull and 0.4 N of friction. Which arrow is the driving force? My example shows ___.	Is the hanging weight always the same as the rope tension on the cart? I would change ___ because ___.

Total moving mass

ES: masa total en movimiento / PT: massa total em movimento

Meaning in my words	How can you tell?
How would you explain total moving mass to a friend? In my words, this means ___.	If you move a washer from the cart onto its hanger, what happens to total moving mass? I can tell because ___.
Example or sketch	**Non-example and why**
A 0.6 kg cart side and 0.1 kg hanger side move together. What total should you use? My example shows ___.	A student adds a new washer from the bench to the hanger. Is moving mass still controlled? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Proportional

ES: proporcional / PT: proporcional

Meaning in my words	How can you tell?
What does it mean to say acceleration is proportional to net force? In my words, this means ___.	At fixed mass, if net force triples, what happens to acceleration? I can tell because ___.
Example or sketch	**Non-example and why**
For a 2 kg cart, compare the accelerations under 2 N and 4 N net. My example shows ___.	Two readings both rise, but one doubles and the other barely changes. Is that enough to call them proportional? I would change ___ because ___.

Friction offset

ES: desfase por fricción / PT: deslocamento devido ao atrito

Meaning in my words	How can you tell?
How would you explain friction offset to a friend? In my words, this means ___.	If the pull is 3 N and opposing friction is 1 N, how much force accelerates the cart? I can tell because ___.
Example or sketch	**Non-example and why**
For a sliding 2 kg cart with 1 N opposing friction, what acceleration would a 3 N pull produce? My example shows ___.	A student uses the full 3 N pull as the net force. What did they leave out? I would change ___ because ___.

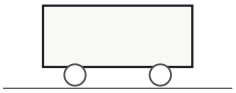
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

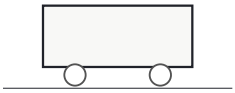
Sketch cart and hanger before and after the transfer. Label the moving mass and the changed driving force.

Labels you can use: driving pull · friction · net force.

Before transfer



After transfer



Tell a partner

The driving pull is ___. After friction, the net force is ___.

Which run changes more?

12

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when net forces differ.

CHOOSE ONE JOB TOGETHER

FLIGHT-TEST ENGINEER ☐

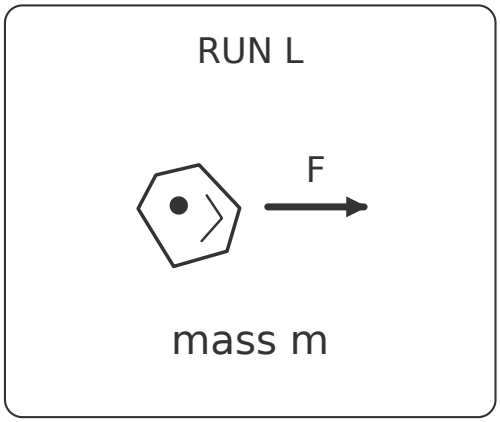
Add velocity-change arrows for both runs. Recommend the larger change and name what stayed the same.

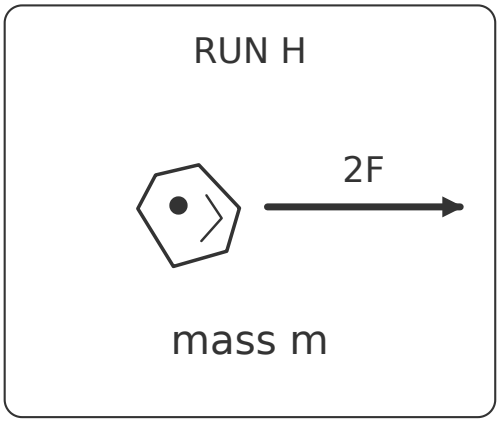
TEST-RANGE BROADCASTER ☐

Tell viewers which run changes velocity more. Ask whether doubling motor pull always doubles the net force.

Word help: net force / drag / velocity change

A rehearsal uses the same asteroid model twice. The second net push is twice the first, for the same time. Operations needs a comparison.





01 OUR FIRST CALL

How do the two velocity changes compare? Point to the force arrows.

The velocity change is ___ if ___ stays the same.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Do the new readings still support your comparison? Explain what you must check.

Now we would say ___ because ___.

Switch speakers. Do those arrows show net force or just motor pull?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks